

Generation Interconnection Request System Impact Study Report

For Queue Project Q24-2428

Pearson Road 100 kV
July 02, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2428. The Interconnection Customer proposes a 120.0 MW solar photovoltaic facility interconnecting at the Pearson Road 100 kV substation in the Illinois area. The request was received on July 08, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 484 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,085,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Lockport / Brown Plant 100 KV Line
Contingency: Kalispell / Milvon 100 KV
Loading: 165.8% (Rating: 120 MVA)
DFAX: 39.2%
Pre-project loading: 126.6%
2. Facility: Kalispell / Milvon 100 KV
Contingency: Locport / Brown Plant 100 KV
Loading: 125.0% (Rating: 148 MVA)
DFAX: 26.2%
Pre-project loading: 103.8%
3. Facility: Lockport / Buckeye Canada 100 KV
Contingency: Lockport / Will County 100 KV
Loading: 104.7% (Rating: 108 MVA)
DFAX: 22.3%
Pre-project loading: 80.0%
4. Facility: Lockport / Will County Substation 100 KV L
Contingency: Buckeye Canada / Addison 100 KV
Loading: 104.3% (Rating: 132 MVA)
DFAX: 9.8%
Pre-project loading: 95.4%
5. Facility: Buckeye Canada / Addison 100 KV L
Contingency: Will County / Lockport 100 KV line
Loading: 103.6% (Rating: 147 MVA)
DFAX: 10.3%
Pre-project loading: 95.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lockport / Brown Plant 100 KV Line	Rebuild line with higher-capacity conductor	\$1,810,000	82% (\$1,475,000)
Kalispell / Milvon 100 KV	Rebuild line with higher-capacity conductor	\$1,950,000	82% (\$1,609,000)
Lockport / Buckeye Canada 100 KV	Reconductor line	\$522,000	81% (\$423,000)
Lockport / Will County Substation 100 KV L	Reconductor line	\$1,014,000	100% (\$1,014,000)
Buckeye Canada / Addison Substation 100 KV	Reconductor line	\$564,000	100% (\$564,000)
	Total allocated to project		\$5,085,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.